

Medical Core · Verified Research

# Mandibular Fracture Reduction Optimization

**Convert an ambiguous clinical problem into a testable 3D registration and optimization problem.**

Period 2021.12 - 2023.02 | Evidence state Verified · Completed | Portfolio tier Medical Core

Built a surgical-planning simulator and experiment pipeline that optimizes mandibular fragment pose using dental features and fracture surfaces.

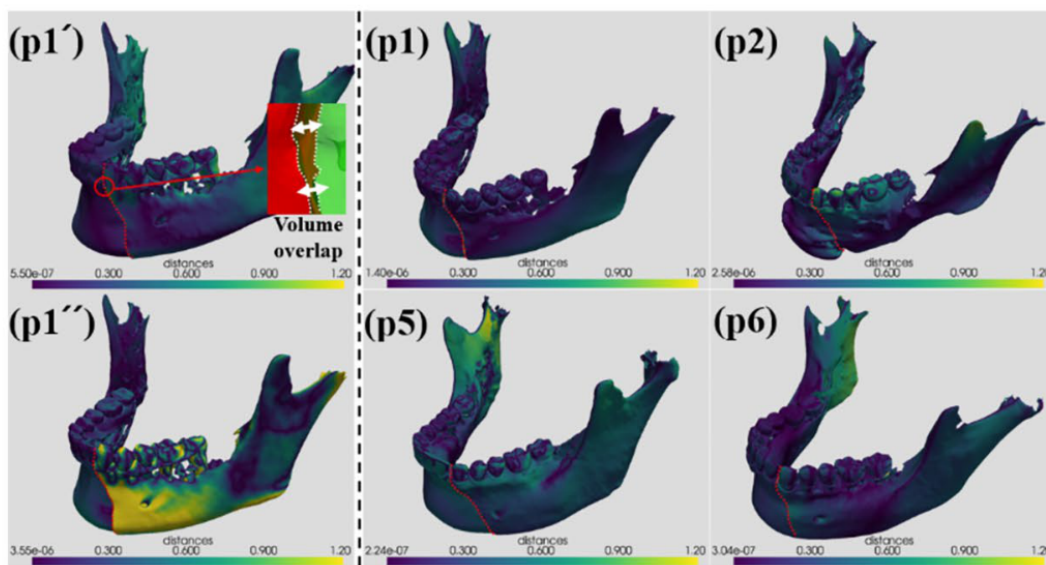


Figure 1. Surface-distance maps between the pre-fracture mandible and the reduction model (part of Fig. 3 in the published paper).

## Problem and system boundary

**Many geometrically plausible reductions existed; the planning target needed a reproducible constraint grounded in occlusion.**

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## Owned decisions and implementation

### My role

Jointly led the research pipeline across problem definition, feature and optimization design, simulator, experiments and analysis, and paper, serving as a co-first author.

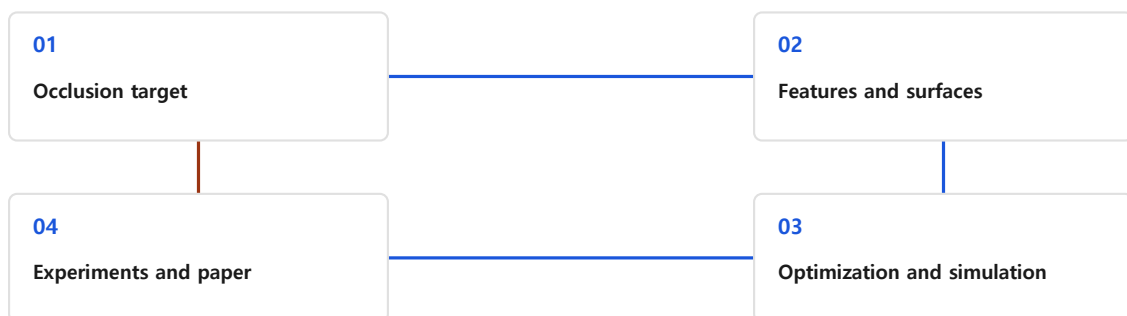
### Problem definition

Defined occlusion as a reproducible target for fracture reduction.

### Research pipeline

- Feature design
- Optimization
- Simulator
- Experiments and analysis
- Paper

### Occlusion-constrained optimization loop



Explanatory system relationship diagram; it is not photographic or experimental evidence.

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## Verification and evidence ledger

ACCAS 2022, a 2022 domestic conference, a 2023 best-paper award, a 2024 Q1 SCIE peer-reviewed paper, and quantitative experiments provide the evidence.

### Published evidence

Separate presentations, award, quantitative experiments, and the peer-reviewed publication.

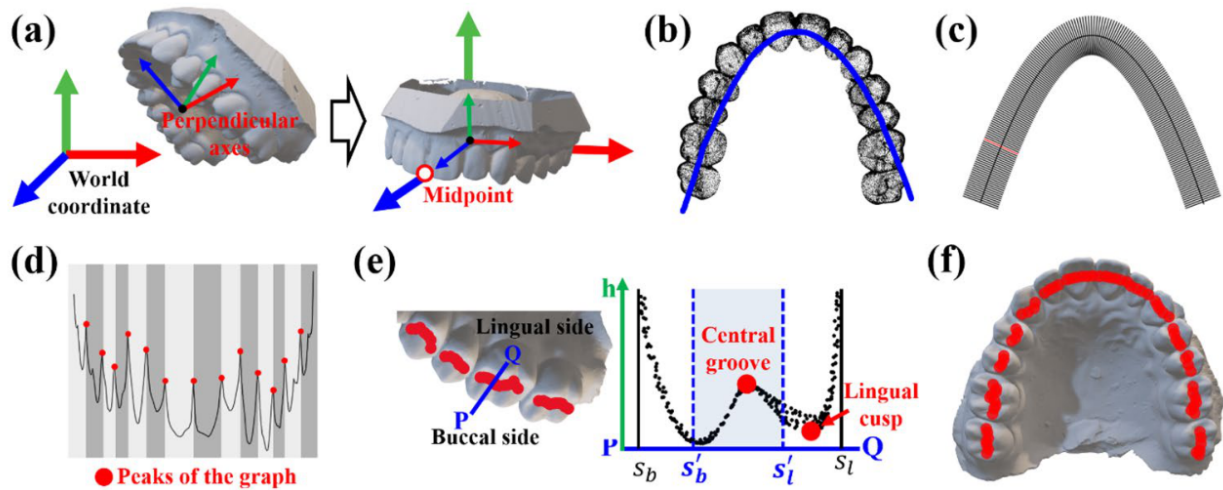


Figure 2. Dental landmark extraction: maxilla alignment, arch-line fit, slicing, central groove, and lingual cusp

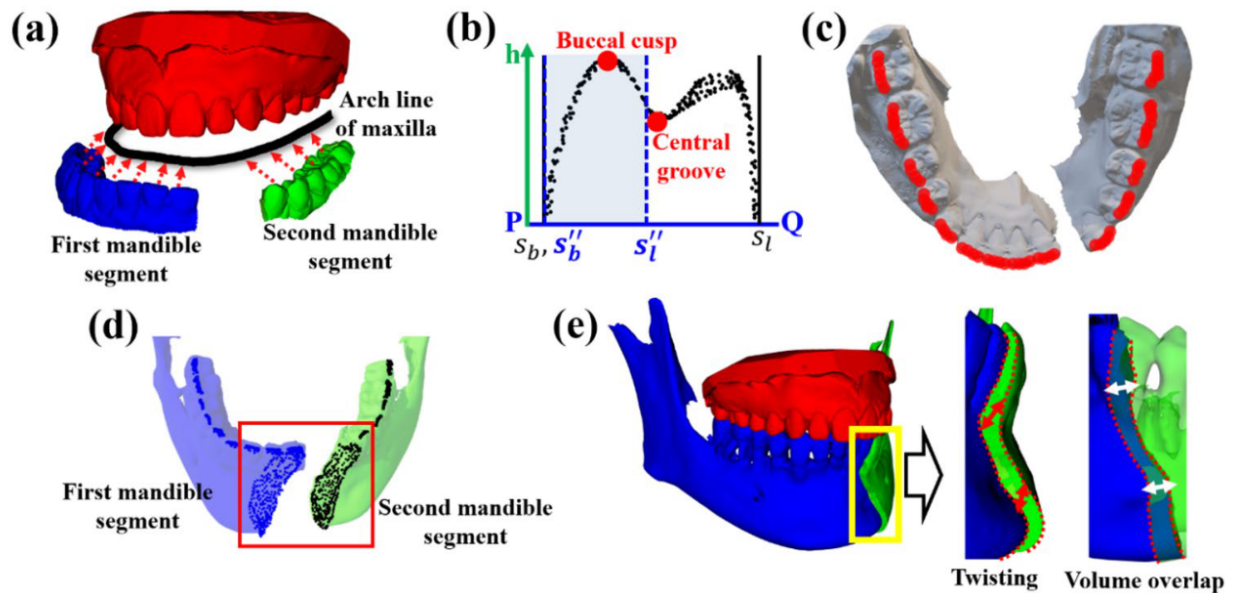


Figure 3. General positioning of fracture segments: arch alignment, fracture-surface features, and overlap error

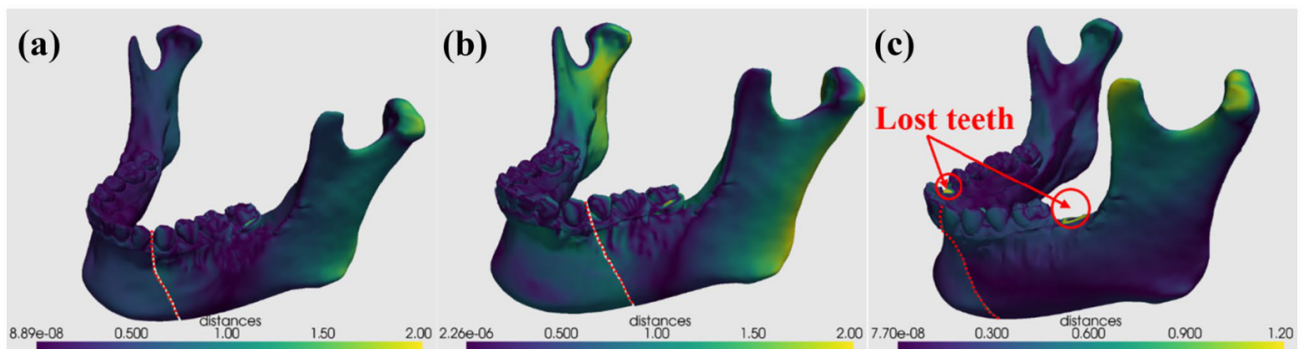


Figure 4. Surface-distance maps for different fracture and tooth conditions: 10-4 split, 11-3 split, two missing teeth

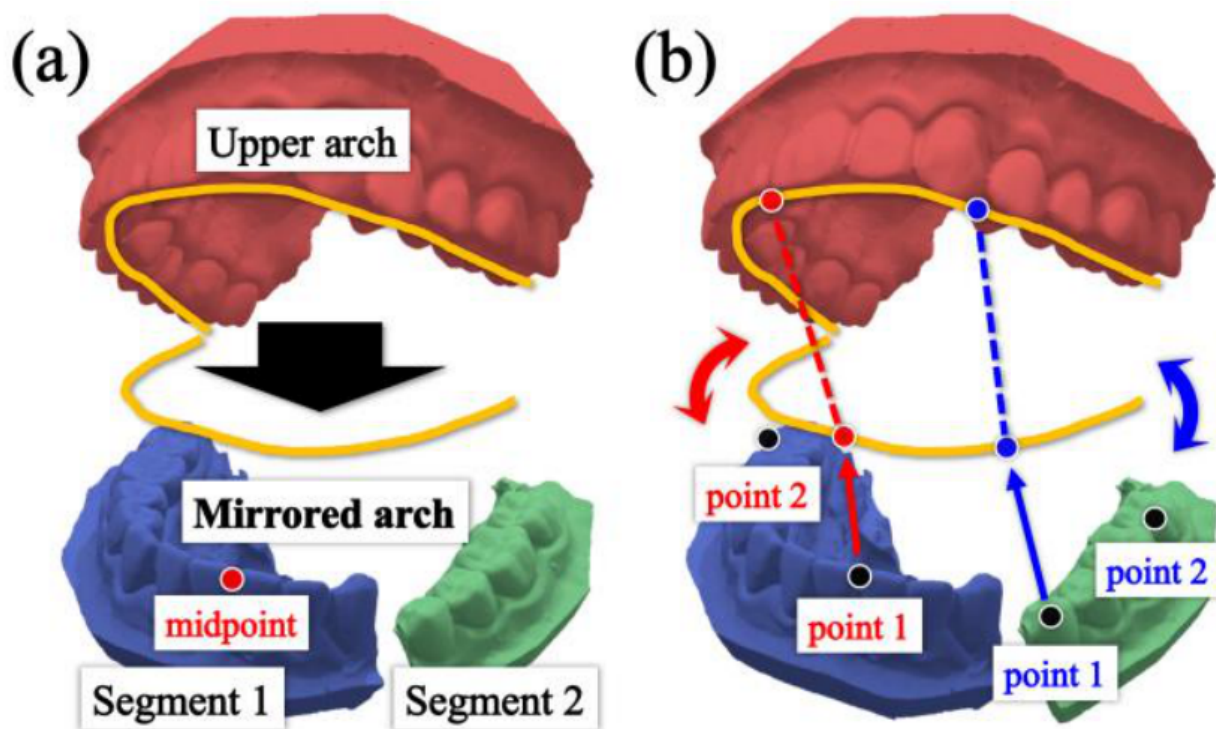


Figure 5. Arch-line-based alignment of mandibular segments (ACCAS 2022)



Figure 6. Presenting at ACCAS 2022, Khon Kaen

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## Role, team result, and limits

### Team result

The joint research team produced international and domestic presentations, an award, and a peer-reviewed publication.

### Limitations

Public evidence is the published figures and presentation material; no clinical use or sole first authorship is claimed.

### Research boundary

Prepared-dataset results do not establish routine clinical use.

### Collaboration boundary

Clinical interpretation, algorithm design, experiment design, and writing were conducted as joint research.

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## Implementation technologies

Python · Open3D · OpenCV · SciPy · PCA · ICP · CLPSO · 3D Slicer

### Primary capability

3D Geometry & Registration · Medical Navigation & Visualization

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## Registered public evidence

[Publication](#)

### What to bring

Share the problem, data or sensors, target validation, and schedule. We can define coordinate and evidence boundaries together.

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